GARSEF
GREATER AUSTIN REGIONAL SCIENCE AND ENGINEERING FAIR
A Program of the Austin Science Education Foundation

JR/SR DIVISION CATEGORIES

ISEF Category descriptions

Animal Sciences (ANIM)
- Agricultural Animal Science
- Animal Behavior
- Cellular Studies
- Development
- Ecology
- Genetics
- Nutrition and Growth
- Physiology
- Systematics & Evolution

Behavioral and Social Sciences (BEHA)
- Behavioral Neuroscience
- Development
- Cognitive Psychology
- Sociology and Anthropology

Biochemistry (BCHM)
- Analytical Biochemistry
- General Biochemistry
- Medicinal Biochemistry
- Structural Biochemistry

Biomedical and Health Sciences (BMED)
- Cell, Organ, and Systems Physiology
- Genetics and Molecular Biology of Disease
- Immunology
- Nutrition and Natural Products
- Pathophysiology

Biomedical Engineering (ENBM) - NEW 2027!
- Biomaterials and Regenerative Medicine
- Biomechanics
- Biomedical Devices
- Biomedical Sensors and Imaging
- Cell and Tissue Engineering
- Synthetic Biology

Cellular and Molecular Biology (CELL)
- Cell Physiology
- Cellular Immunology
- Genetics
- Molecular Biology
- Neurobiology

Chemistry (CHEM)
- Analytical Chemistry
- Computational Chemistry
- Environmental Chemistry
- Organic/Inorganic Chemistry
- Materials Chemistry
- Physical Chemistry

Computational Biology and Bioinformatics (CBIO) - NEW 2027 - both divisions!
- Computational Biomodelling
- Computational Epidemiology
- Computational Evolutionary Biology
- Computational Neuroscience
- Computational Pharmacology
- Genomics

Earth and Environmental Sciences (EAEV)
- Atmospheric Science
- Climate Science
- Environmental Effects on Ecosystems
- Geosciences
- Water Science

Embedded Systems (EBED)
- System Hardware
- Circuits
- Internet of Things
- Microcontrollers
- Networking/Communications
- Optics
- Sensors
- Signal Processing
- Robotic Mechanical Design

Energy: Sustainable Materials and Design (EGSD)
- Biological Process and Design
- Solar Process, Materials, and Design
- Energy Storage
- Wind and Water Movement Power Generation
- Hydrogen Generation and Storage
- Thermal Generation and Design
- Triboelectricity and Electrolysis

Engineering Technology: Statics and Dynamics (ETSD)
- Aerospace and Aeronautical Engineering
- Civil Engineering
- Computational Mechanics
- Control Theory
- Electrical Engineering
- Ground Vehicle Systems
- Industrial Engineering-Processing
- Mechanical Engineering
- Naval Systems
- Structural Engineering

Environmental Engineering (ENEV)
- Bioremediation
- Land Reclamation
- Pollution Control
- Recycling and Waste Management
- Water Resources Management

Materials Science (MATS)
- Biomaterials
- Ceramic and Glasses
- Composite Materials
- Computation and Theory
- Electronic, Optical, and Magnetic Materials
- Nanomaterials
- Polymers

Mathematics (MATH)
- Algebra
- Combinatorics, Graph Theory, and Game Theory
- Geometry and Topology
- Mathematical Analysis
- Number Theory
- Probability and Statistics

Microbiology (MCRO)
- Antimicrobials and Antibiotics
- Applied Microbiology
- Bacteriology
- Environmental Microbiology
- Microbial Genetics
- Virology

Physics and Astronomy (PHYS)
- Atomic, Molecular, and Optical Physics
- Astronomy and Cosmology
- Biological Physics
- Condensed Matter and Materials
- Mechanics
- Nuclear and Particle Physics
- Theoretical, Computational, and Quantum Physics

Plant Sciences (PLNT)
- Agriculture and Agronomy
- Ecology
- Genetics and Breeding
- Growth and Development
- Pathology
- Plant Physiology
- Systematics and Evolution

Software Design (SFTD)
- Algorithms
- Cybersecurity
- Databases
- Human/Machine Interface
- Languages and Operating Systems
- Mobile Apps
- Robotics Artificial Intelligence/Cognitive Systems
- Machine Learning

Translational Medical Sciences (aka Disease Studies) (TMED)
- Disease Detection and Diagnosis
- Disease Prevention
- Disease Treatment and Therapies
- Drug Identification and Testing
- Pre-Clinical Studies